A Sample Lecture on Graph Search

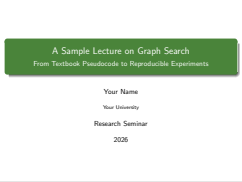
From Textbook Pseudocode to Reproducible Experiments

Your Name

Your University

Research Seminar

2026



Hi everyone. **Thanks for being here.** Today I am walking through a sample lecture on graph search.

We will explore where the classic algorithms come from and what they mean for real workloads.

The big idea is: if we want results people can trust, we need structure, not just speed.

Before diving in, let me quickly outline what we will cover today.

Content

1 Introduction
2 Related Work
3 Approach
4 Foundations
5 Evaluation
6 Conclusion

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Six parts. Spend most of the time on the approach and the evaluation. Skip the foundations if the audience already knows BFS.

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- Road networks, citation graphs and dependency trees

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Reveal one bullet at a time. Ask who has tried to reproduce a benchmark.

- Road networks: mention the city map example
- **Pause** before the last point

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Both questions are answered in the evaluation section. Point forward to the summary table.

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A Short History

- Shortest paths with a priority queue [1]
- Heuristic search with admissible estimates [2]
- Community benchmarks for reproducibility [3]

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The citations are links. In TeXora, hover one to peek at the reference without leaving this slide.

The Core Recurrence

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Search Pipeline



Search Pipeline
Graph

Build the diagram step by step. The loop back is the part people forget.

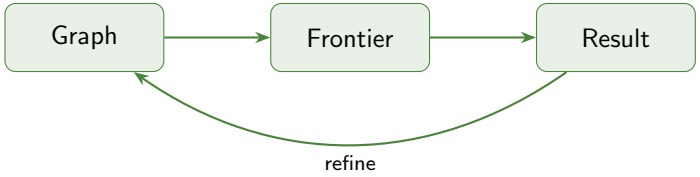
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Search Pipeline
Graph
Frontier

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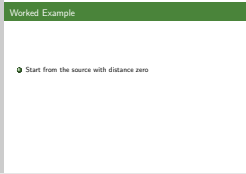




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Worked Example

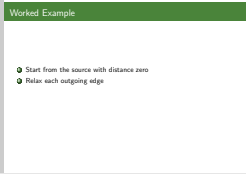
1. Start from the source with distance zero



Use the pen to trace the order on the board. Use the laser to point at step three.

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Complexity at a Glance

Method	Time	Space
Breadth-first	$O(V + E)$	$O(V)$
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Mention that the theoretical winner is rarely the practical winner.

Experimental Setup

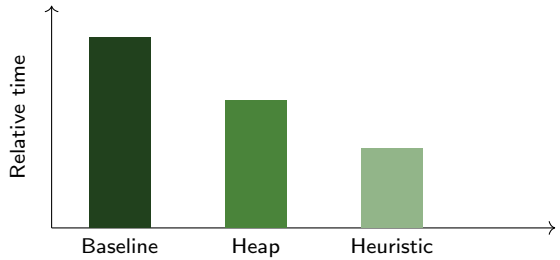
- Twelve sparse graphs, three dense graphs
- Five repetitions, median reported
- Fixed seeds and pinned library versions

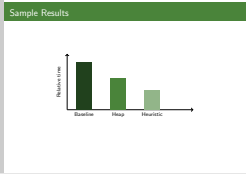
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Keep this short. Details are in the backup slide if someone asks.

Backup: full setup.

Sample Results





These numbers are invented placeholders for the demo. The takeaway: the heuristic halves the runtime.

Take-Home Messages

1. Choose the frontier for your graph, not for the textbook
2. Publish seeds, versions and raw data
3. Questions welcome

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Pause here. Leave this slide up during the questions.

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- Compiler optimisation level three
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Only show this if someone asks about the hardware.

References

[1] E. W. Dijkstra. A note on two problems in connexion with graphs. *Numerische Mathematik*, 1959.

[2] P. E. Hart, N. J. Nilsson and B. Raphael. A formal basis for the heuristic determination of minimum cost paths. *IEEE Trans. SSC*, 1968.

[3] Placeholder Authors. Sample benchmark collection. *Example Proceedings*, 2020.

References

[1] E. W. Dijkstra. A note on two problems in connexion with graphs. Numerische Mathematik, 1959.

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Reference slide. Reached from the citation links.